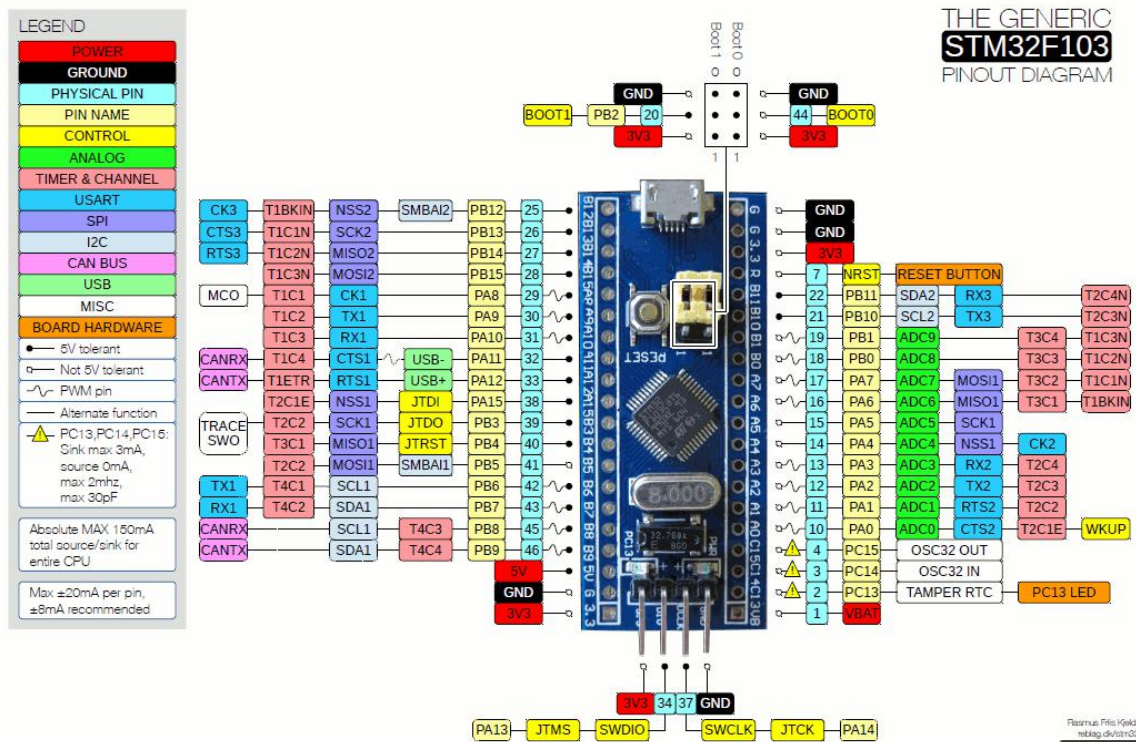
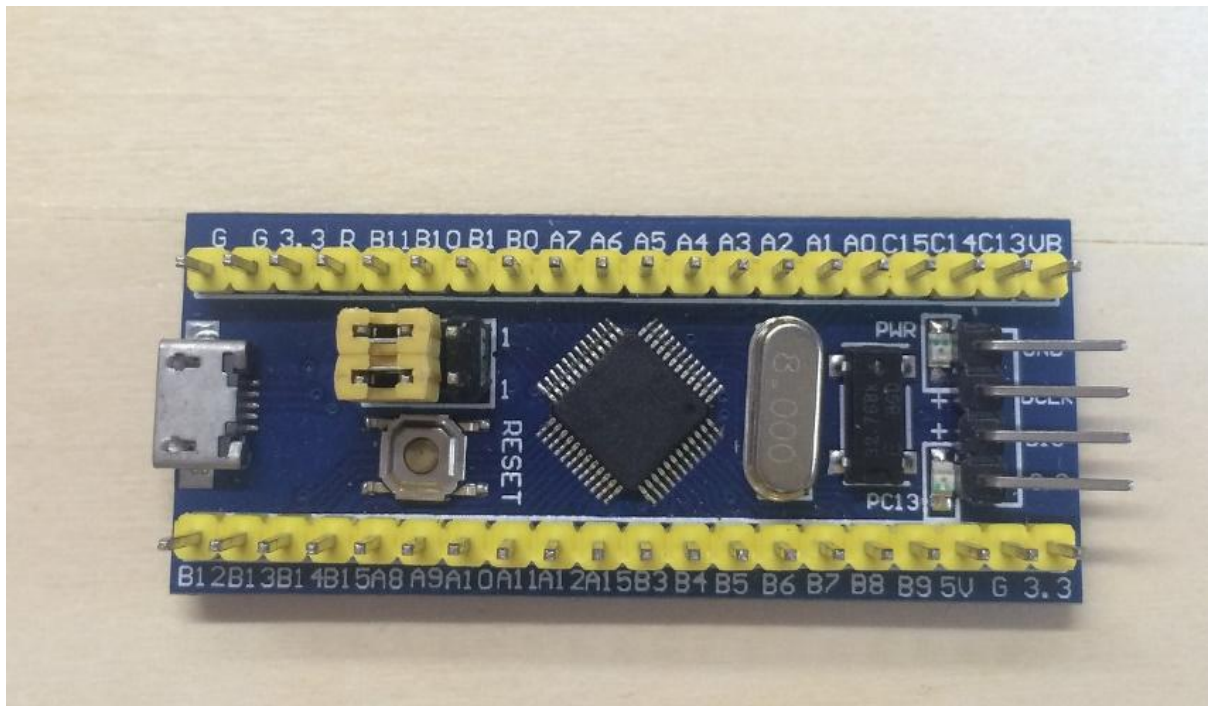


STM32Duino 的引脚映射，ADC，PWM 等介绍



0、编程语法参考

<http://docs.leaflabs.com/static.leaflabs.com/pub/leaflabs/maple-docs/latest/language.html>

1、引脚对应关系：

32 个 GPIO，也可以用序号表达。

28	B12	STM32 开发板	GND	
29	B13		GND	
30	B14		3.3v	
31	B15		R	
8	PA8		PB1	27
9	PA9		PB1	26
10	A10		PB1	17
11	A11		PB0	16
12	A12		PA7	7
15	A15		PA6	6
19	PB3		PA5	5
20	PB4		PA4	4
21	PB5		PA3	3
22	PB6		PA2	2
23	PB7		PA1	1
24	PB8		PA0	0
25	PB9		PC1	32
5v			PC1	33
GND			PC1	34
3.3v			VBAT	

序号 物理引脚

- 0 PA0
- 1 PA1
- 2 PA2
- 3 PA3
- 4 PA4
- 5 PA5
- 6 PA6
- 7 PA7
- 8 PA8

9	PA9
10	PA10
11	PA11
12	PA12
13	PA13
14	PA14
15	PA15
16	PB0
17	PB1
18	PB2
19	PB3
20	PB4
21	PB5
22	PB6
23	PB7
24	PB8
25	PB9
26	PB10
27	PB11
28	PB12
29	PB13
30	PB14
31	PB15
32	PC13
33	PC14
34	PC15

2、允许 PWM 引脚：

序号	物理引脚
0	PA0
1	PA1
2	PA2
3	PA3
6	PA6
7	PA7
8	PA8
9	PA9
10	PA10
16	PB0
22	PB6
23	PB7

PWM 引脚简易测试程序，总共有 12 个 PWM 引脚，16bit 即 65536 级 PWM，很精密了：

```
1. void setup()
2. {
```

```

3.         pinMode(PB0, PWM);
4.         pinMode(PA7, PWM);
5.         pinMode(PA6, PWM);
6.         pinMode(PA3, PWM);
7.         pinMode(PA2, PWM);
8.         pinMode(PA1, PWM);
9.         pinMode(PA0, PWM);
10.        pinMode(PB7, PWM);
11.        pinMode(PB6, PWM);
12.        pinMode(PA10, PWM);
13.        pinMode(PA9, PWM);
14.        pinMode(PA8, PWM);
15.    }
16.    void loop()
17.    {
18.        for (int i = 0; i < 65536; i++)
19.        {
20.            delayMicroseconds(40);
21.            pwmWrite(PB0, i);
22.            pwmWrite(PA7, i);
23.            pwmWrite(PA6, i);
24.            pwmWrite(PA3, i);
25.            pwmWrite(PA2, i);
26.            pwmWrite(PA1, i);
27.            pwmWrite(PA0, i);
28.            pwmWrite(PB7, i);
29.            pwmWrite(PB6, i);
30.            pwmWrite(PA10, i);
31.            pwmWrite(PA9, i);
32.            pwmWrite(PA8, i);
33.        }
34.    }

```

复制代码

PWM 精度太高了，驱动 LED 没必要那么高，再来一个指数式适合驱动 LED 的，看起来亮度更线性。

```

1.    void setup() {
2.        pinMode(PB0, PWM);
3.        pinMode(PA7, PWM);
4.        pinMode(PA6, PWM);
5.        pinMode(PA3, PWM);
6.        pinMode(PA2, PWM);
7.        pinMode(PA1, PWM);
8.        pinMode(PA0, PWM);
9.        pinMode(PB7, PWM);
10.       pinMode(PB6, PWM);

```

```

11.     pinMode(PA10, PWM);
12.     pinMode(PA9, PWM);
13.     pinMode(PA8, PWM);
14. }
15. void loop() {
16.     //指数式增加亮度, 适合驱动 LED, 看起来亮度更线性
17.     for (int i = 0; i < 256; i++)
18.     {
19.         delay(20);
20.         pwmWrite(PB0, i * i);
21.         pwmWrite(PA7, i * i);
22.         pwmWrite(PA6, i * i);
23.         pwmWrite(PA3, i * i);
24.         pwmWrite(PA2, i * i);
25.         pwmWrite(PA1, i * i);
26.         pwmWrite(PA0, i * i);
27.         pwmWrite(PB7, i * i);
28.         pwmWrite(PB6, i * i);
29.         pwmWrite(PA10, i * i);
30.         pwmWrite(PA9, i * i);
31.         pwmWrite(PA8, i * i);
32.     }
33. }

```

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或者用序号表示:

```

1.     int pins[12] = {0, 1, 2, 3, 6, 7, 8, 9, 10, 16, 22, 23};
2.     void setup()
3.     {
4.         for (int i = 0; i < 12; i++)
5.         {
6.             pinMode(pins[i], PWM);
7.         }
8.     }
9.     void loop()
10.    {
11.        for (int i = 0; i < 100; i++)
12.        {
13.            for (int j = 0; j < 12; j++)
14.            {
15.                pwmWrite(pins[j], i * i);
16.            }
17.            delay(20);
18.        }
19.    }

```

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3、允许的 ADC 引脚：

ADC 总共有 10 个通道，12bit 即 4096 级。引脚如下

物理引脚，ADC 通道

PA0	CH0
PA1	CH1
PA2	CH2
PA3	CH3
PA4	CH4
PA5	CH5
PA6	CH6
PA7	CH7
PB0	-(不能用序号表达)
PB1	-(不能用序号表达)

ADC 测试程序：

```
1. void setup()
2. {
3.     Serial.begin(115200);
4.     pinMode(PB0, INPUT_ANALOG);
5.     pinMode(PA7, INPUT_ANALOG);
6.     pinMode(PA6, INPUT_ANALOG);
7.     pinMode(PA5, INPUT_ANALOG);
8.     pinMode(PA4, INPUT_ANALOG);
9.     pinMode(PA3, INPUT_ANALOG);
10.    pinMode(PA2, INPUT_ANALOG);
11.    pinMode(PA1, INPUT_ANALOG);
12.    pinMode(PA0, INPUT_ANALOG);
13.    pinMode(PB1, INPUT_ANALOG);
14. }
15. void loop()
16. {
17.     delay(50);
18.     Serial.print("\tPB0="); Serial.print(analogRead(PB0));
19.     Serial.print("\tPA7="); Serial.print(analogRead(PA7));
20.     Serial.print("\tPA6="); Serial.print(analogRead(PA6));
21.     Serial.print("\tPA5="); Serial.print(analogRead(PA5));
22.     Serial.print("\tPA4="); Serial.print(analogRead(PA4));
23.     Serial.print("\tPA3="); Serial.print(analogRead(PA3));
24.     Serial.print("\tPA2="); Serial.print(analogRead(PA2));
25.     Serial.print("\tPA1="); Serial.print(analogRead(PA1));
26.     Serial.print("\tPA0="); Serial.print(analogRead(PA0));
27.     Serial.print("\tPB1="); Serial.println(analogRead(PB1));
28. }
```

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4、允许的串口

总共 3 个硬件外接串口 Serial1、Serial2、Serial3，一个 USB 虚拟串口 Serial:

```
1. void setup()
2. {
3.     Serial.begin(115200);
4.     Serial1.begin(115200); //TX=PA9,RX=PA10
5.     Serial2.begin(115200); //TX=PA2,RX=PA3
6.     Serial3.begin(115200); //TX=PB10,RX=PB11
7. }
8. void loop()
9. {
10.    delay(100);
11.    Serial.println("Test Serial");
12.    Serial1.println("Test Serial1");
13.    Serial2.println("Test Serial2");
14.    Serial3.println("Test Serial3");
15. }
```

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